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83%.

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **CSA16 COURSE CODE: Data Warehousing and Data Mining**

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:20**

Annexure A
SHORT ANSWER TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	*Total(10)*
Weightage	*30%*	25%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I K. Yoganand (192525173) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Yoganand

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1 Data warehouse (Time, Doctor, patient)

(a) three schema types :-

* star schema
* Snowflake schema
* fact constellation (galaxy) schema

(b) star schema

Time
|
Doctor ---- fact (fee) --- patient
|
Measures:
count, charge

(c) SQL Query

```
SELECT day, month, year, doctor, hospital, patient
       SUM (count) AS total_count,
       SUM (charge) AS total_charge
FROM fee
GROUP BY day, month, year, doctor,
   hospital, patient;
```

2 By university data warehousing

(a) Snowflake schema

student
|
Course - - - fact Table - - - semester
|
Instructor

Measures;

* Count
* Avg - Grade

(b). OLAP Operations

1. Roll-up semester → year
2. slice Course = 'CSC'
3. Roll-up Instructor → All
4. Roll up -student → student

Result :- Average grade of CSC Courses for each student

(c) Number of Cuboids

Each dimension has 4 levels

formula;

Number of cuboids

$= 4^4 = 256$

... General Data warehousing Design.

Dimensions!

* Time (hour → Day → month → year)
* Location (state - city - Region)
* weather type

Measures!

* Air pressure
* Temperature
* precipitation

schema! star schema

this design supports efficient OLAP operations such as roll-up, drill down, slice and dice

4 Data cube (n dimensions, each with p values)

(a) maximum base cuboid cells

p^n

(b) minimum base cuboid cells

0

(If no data exists)

(c) maximum cells in data cube

$(p+1)^n$

(d) minimum cells in the data cube

1

(the apex cuboid)

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4

(a) the base cuboid is the most detailed level of the data cube, every combination of the value in all dimensions creates one cell

* Number of dimensions = n
* Distinct value in each dimension = p

p^n Ex:- $n=3, p=4$ $= 4^3 = 64$

(b) the minimum occurs when no data records are present in the cube

∴, if the database is empty, there are 0 cells

(c) A complete data cube includes

* the base cuboid
* all aggregate cuboids
* the apex cuboid (All dimensions)

$\therefore (p+1)^n$ Ex:- $n=3, p=4$ $= (4+1)^3 = 5^3 = 125$

(d) the smallest possible data cube contains only the apex cuboid, which represents the aggregation of all dimensions.